



EL2805 Reinforcement Learning

Exercise Session 6

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Instructions.

- These exercises are for your practice, and you do not need to submit solutions. We encourage you to work on them before the exercise session on Dec 11th.
- In each exercise session, we will go through a selection of topics chosen by the teaching assistant and participating students.
- Homework and labs will be based on concepts covered in lectures and exercise sessions. These materials are excellent preparation for your graded assignments!
- There is no need to memorize equations - relevant materials will be provided during the exam. Focus on concepts and understanding!
- The materials cover a wide range of topics, so feel free to focus on what interests you most. However, to develop a well-rounded understanding of RL, it is important that you engage with both theoretical and practical exercises.
- We use green boxes to highlight established results and definitions, yellow for problems that require your solutions, and pink for high-level remarks. Exercises marked with a ★ are more challenging and intended for students seeking an extra challenge.
- We value your feedback on the materials provided. If you have any questions, concerns, or suggestions, do not hesitate to reach out to any of the TAs.

LLM Alignment: Preferences, Rewards, and a Dash of DPO

For some tasks, defining explicit rewards can be challenging (just think about making scrambled eggs!). While we might attempt to design a simple reward function that aligns with our objectives, it often risks misrepresenting our true intentions. Conversely, methods like inverse reinforcement learning, which infer rewards from demonstrations, are not feasible when suitable demonstrations are unavailable—such as when we lack access to the sequence of states and actions a robot would experience while making scrambled eggs.

As a middle ground, Reinforcement Learning from Human Feedback (RLHF) [1] has emerged as a powerful approach for aligning machine learning models with complex, hard-to-specify goals. In RLHF, feedback is provided as binary preferences, a concept we will explore in this session.

RLHF has gained prominence as a foundational technique in training state-of-the-art Large Language Models (LLMs), including OpenAI’s GPT-4, Anthropic’s Claude, Google’s Bard, and Meta’s Llama 2-Chat. In this exercise session, we will delve into RLHF and the state-of-the-art algorithm Direct Preference Optimization (DPO), focusing on their roles in fine-tuning LLMs. Before diving deeper, consider the following question:

Exercise 1. Explain why reinforcement learning is not used for full training of LLMs. What would the underlying MDP look like? How does the fine-tuning stage differ?

1.1 Learning the Preferences

As a warm-up, we will consider the task of learning a model of preferences. Suppose we are given a dataset $\mathcal{D}$ of the form $\{(y_w^i, y_\ell^i)\}_{i=1}^{|\mathcal{D}|}$, where y_w ’s are preferred responses (w for winners) over y_ℓ ’s (ℓ for losers). Our goal is to learn a model that predicts the probability $P(y \succ z)$, representing the likelihood that y is preferred over z . This is achieved by minimizing the following loss function:

$$\min_P \mathcal{L}_P(\mathcal{D}) = \min_P -\mathbb{E}_{(y_w, y_\ell) \sim \mathcal{D}} [\log P(y_w \succ y_\ell)] \approx \min_P \left[-\sum_{i=1}^{|\mathcal{D}|} \log P(y_w^i \succ y_\ell^i) \right]$$

Exercise 2. You want to learn a preference model for Swedish pastries. You conduct a survey asking people whether they prefer *kanelbulle* (ℓ) over *kardemummabulle* (m), and you collect 25 responses of the form (ℓ, m) (indicating preference for *kanelbulle*) and 10 of the form (m, ℓ) . What probability model would minimizing the loss above yield?

1.2 RLHF: Model the Reward, then Optimize

Most RLHF algorithms consist of two separate, but interconnected, stages: reward modeling and policy optimization. Let’s discuss both of them in detail:

Stage I: Reward Modeling

In the RL framework, preferences can be expressed as the preferred choice of action in a particular state. We consider a dataset $\mathcal{D} = \{s^i, a_w^i, a_\ell^i\}_{i=1}^{|\mathcal{D}|}$, where s^i are the prompts/states, a_w^i are the preferred responses/actions, and a_ℓ^i are the dispreferred responses/actions. To transition from probabilities to preference-based rewards, we employ the Bradley-Terry model (dating back to 1952!), which assumes the following:

$$P(a_w \succ a_\ell | s) = \sigma(r(s, a_w) - r(s, a_\ell)) = \frac{\exp(r(s, a_w))}{\exp(r(s, a_w)) + \exp(r(s, a_\ell))}$$

where σ is the sigmoid function, defined as $\sigma(x) = \frac{1}{1 + \exp(-x)}$.

Exercise 3. Repeat Exercise 2, but this time suggest rewards $r(m)$ and $r(\ell)$ that model preferences of surveyed individuals according to the Bradley-Terry model. Are the rewards uniquely defined?

In practice, we parameterize the reward function using a neural network with parameters ϕ i.e., let $r_\phi(s, a)$. We then combine Bradley-Terry model with cross-entropy loss $\mathcal{L}_P$, introduced earlier. Minimizing this loss provides optimal parameters ϕ . Concisely, the learned reward model can be expressed as the solution to:

$$\min_{\phi} \mathcal{L}_r(\phi, \mathcal{D}) = \min_{\phi} -\mathbb{E}_{(s, a_w, a_\ell) \sim \mathcal{D}} \left[\log \sigma \left(r_\phi(s, a_w) - r_\phi(s, a_\ell) \right) \right]$$

Exercise 4. Human feedback for every time step and each action may be difficult to collect. Instead, human evaluators provide preferences for trajectories τ of a fixed length T , ranking them as $\tau^i \succ \tau^j$ if trajectory τ^i is preferred over τ^j . How would you use the Bradley-Terry model to learn rewards from such trajectory-based preferences?

Stage II: Policy Optimization

Before applying RLHF, LLMs are typically trained using supervised learning, including one stage of Supervised Fine-Tuning (SFT). Thus, our starting point is a model fine-tuned with supervised learning, that we denote as the reference model $\pi_{\text{ref}} = \pi_{\text{SFT}}$.

Now, once we have obtained the reward model, we return to a familiar RL framework. To fine-tune the model while maximizing the learned reward, we aim to optimize the following objective:

$$\max_{\theta} \mathcal{L}_{\pi}(\theta, \mathcal{D}) = \max_{\theta} \mathbb{E}_{s \sim \mathcal{D}} \left[\mathbb{E}_{a \sim \pi_{\theta}(\cdot|s)} [r_{\phi}(s, a)] - \beta D_{\text{KL}}(\pi_{\theta}(\cdot|s) || \pi_{\text{ref}}(\cdot|s)) \right]$$

where β is a regularization parameter that controls the extent of change allowed in the policy relative to the reference model.

Exercise 5. Propose a reward proxy $r(s, a)$ to replace the reward using the objective above. Suggest how PPO could be used to optimize the policy π_{θ} . Explain the difference between π_{ref} and π_{old} in the PPO objective. Is it problematic that PPO is an on-policy algorithm?

Beware of what you are optimizing.

When learning the policy, we rely on the learned reward model r_{ϕ} , which is trained on the dataset $\mathcal{D}$. Keeping this in mind, consider the following exercise:

Exercise 6. The Figure on the left reproduces Figure 5 from [7]. Explain what this figure illustrates. This phenomenon is known as reward hacking - what do you think why? Does the figure suggest that fine-tuning improves performance compared to the reference model? How do you think the summaries generated by this overtrained LLM might look? (you can find them in Table 29 in [7]!)

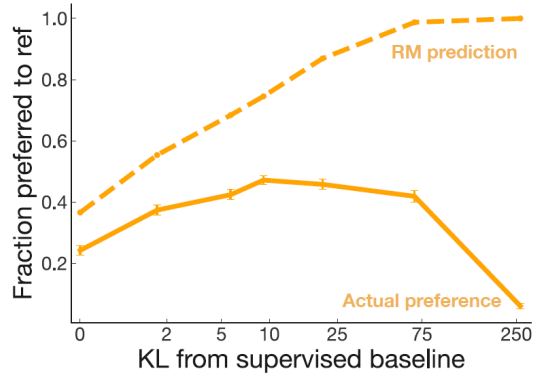


Figure 1: Figure 5 from [7]. RM refers to the reward model. The x -axis shows $D_{\text{KL}}(\pi_{\theta}(\cdot|s) || \pi_{\text{ref}}(\cdot|s))$ for different policies π_{θ} , while the y -axis indicates the proportion of times the fine-tuned model's responses are preferred over the reference model.

1.3 DPO: Don't Model, just Optimize

The genesis of Direct Preference Optimization (DPO)[6] came from addressing several challenges in RLHF: 1. the method was complex and unstable, 2. it required careful tuning of the reward model, and 3. the training process was computationally expensive and often difficult to replicate.

DPO has gained significant attention for its ability to replace RLHF in fine-tuning LLMs without requiring RL - it relies solely on supervised learning. We will now derive the loss function of this algorithm. First, note that the loss function $\mathcal{L}_\pi(\theta, \mathcal{D})$ from the previous section can be expressed as $\mathcal{L}_\pi(\theta, \mathcal{D}) = \mathbb{E}_{s \sim \mathcal{D}}[\mathcal{L}_\pi^s(\theta, \mathcal{D})]$.

Exercise 7. Fix a state s . Consider optimizing $\max_\theta \mathcal{L}_\pi^s(\theta, \mathcal{D})$ under the constraint that $\sum_a \pi_\theta(a|s) = 1$. Solve this optimization problem using Lagrange multipliers and show that the optimal solution is:

$$\pi_\theta^*(a|s) = \frac{1}{Z(\beta, s)} \pi_{\text{ref}}(a|s) \exp\left(\frac{1}{\beta} r_\phi(s, a)\right)$$

where $Z(\beta, s)$ is the partition function given by: $Z(\beta, s) := \sum_a \pi_{\text{ref}}(a|s) \exp\left(\frac{1}{\beta} r_\phi(s, a)\right)$.

From the optimal policy derived in Exercise 7, we can express the rewards corresponding to this policy as:

$$r_\phi^*(s, a) = \beta \log \frac{\pi_\theta(a|s) Z(\beta, s)}{\pi_{\text{ref}}(a|s)} = \beta \log \frac{\pi_\theta(a|s)}{\pi_{\text{ref}}(a|s)} + \beta \log Z(\beta, s)$$

Substituting this reward model into the loss $\mathcal{L}_r(\phi, \mathcal{D})$ from Stage I of RLHF, we obtain the following optimization problem:

$$\begin{aligned} \min_\phi \mathcal{L}_r(\phi, \mathcal{D}) &= \min_\phi -\mathbb{E}_{(s, a_w, a_\ell) \sim \mathcal{D}} \left[\log \sigma \left(r_\phi^*(s, a_w) - r_\phi^*(s, a_\ell) \right) \right] \\ &= \min_\theta -\mathbb{E}_{(s, a_w, a_\ell) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(a_w|s)}{\pi_{\text{ref}}(a_w|s)} - \beta \log \frac{\pi_\theta(a_\ell|s)}{\pi_{\text{ref}}(a_\ell|s)} \right) \right] =: \min_\theta \mathcal{L}_{\text{DPO}}(\theta, \mathcal{D}) \end{aligned}$$

In this way, DPO completely skips the reward modeling step and directly optimizes the policy π_θ using the preference dataset $\mathcal{D}$. Note also that this optimization problem is similar to regression in supervised learning and, as a result, does not require any RL machinery.

Exercise 8. Consider the setting of Exercise 2. This time, suppose that $P(\ell \succ m) = 0.9$ and $P(m \succ \ell) = 0.1$. On a cold December morning, you manage to survey only two people, and both prefer ℓ . What would the optimal DPO-policy be in this case? Does it depend on the regularization parameter β ? How would this change if using RLHF?

Can RL be used for other tasks in LLMs beyond fine-tuning? The role of RL in training LLMs is still not entirely determined. While the DPO vs PPO-based RLHF debate continues, there are emerging approaches using RL in novel ways, such as o1 from OpenAI (<https://openai.com/index/learning-to-reason-with-llms/>), and TULU [4].

Do LLMs speak Swedish? GPT-SW3 [2, 3] is an open-source model trained on 100GB of Swedish (or, rather, 1.2TB of Nordic languages [5]) data. Different models are available at <https://huggingface.co/AI-Sweden-Models>. The two biggest challenges identified during the training were: 1. access to sufficient amounts of diverse, high-quality data, and 2. access to sufficient computational resources. According to the authors: "training process took 2.5 days utilizing 16 of the DGX A100 servers (in total 128 GPUs)." This illustrates that training models from scratch is beyond reach for most individuals or companies. However, fine-tuning is far more feasible, requiring neither such large datasets nor extensive computational power. For more information about GPT-SW3, visit: <https://www.ai.se/en/project/gpt-sw3>.

References

- [1] Paul F Christiano et al. “Deep reinforcement learning from human preferences”. In: *Advances in neural information processing systems* 30 (2017).
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